INNOVATIVE MINING SYSTEMS
Long wave of technology

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Introduction

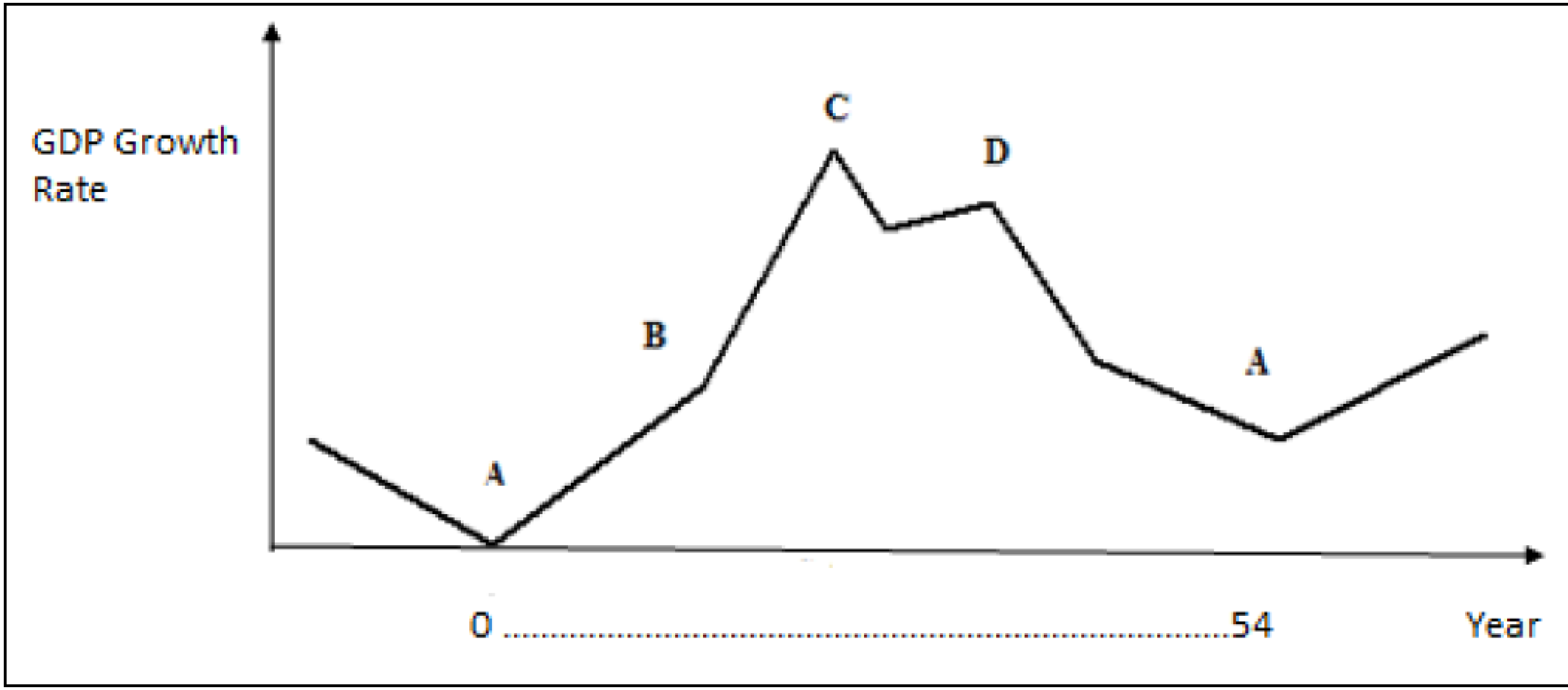
Long Wave Economic Theory explains long-term cyclical movements in economic growth observed in capitalist economies. Proposed by Nikolai D. Kondratieff in the 1920s, the theory states that economic development does not occur at a constant or linear rate, but through long waves of expansion and contraction lasting about 45–60 years.

These long waves, also known as Kondratieff Waves, are identified using long-term data on prices, production, wages, interest rates, and trade. Each wave consists of four phases: recovery, expansion (prosperity), recession, and depression. Together, these phases form a repeating pattern of long-term economic rise and decline.

A key explanation of long waves was given by Joseph Schumpeter, who emphasized the role of technological innovations. According to him, major innovations occur in clusters and lead to the creation of new industries while making older ones obsolete—a process known as creative destruction. Examples include the steam engine, railways, electricity, automobiles, and information technology, each associated with a separate long wave.

Thus, Long Wave Economic Theory highlights the strong link between technological revolutions, structural economic change, and long-term economic growth, showing that sustained development is driven by periodic waves of innovation rather than continuous gradual progress.

Structure of the long wave



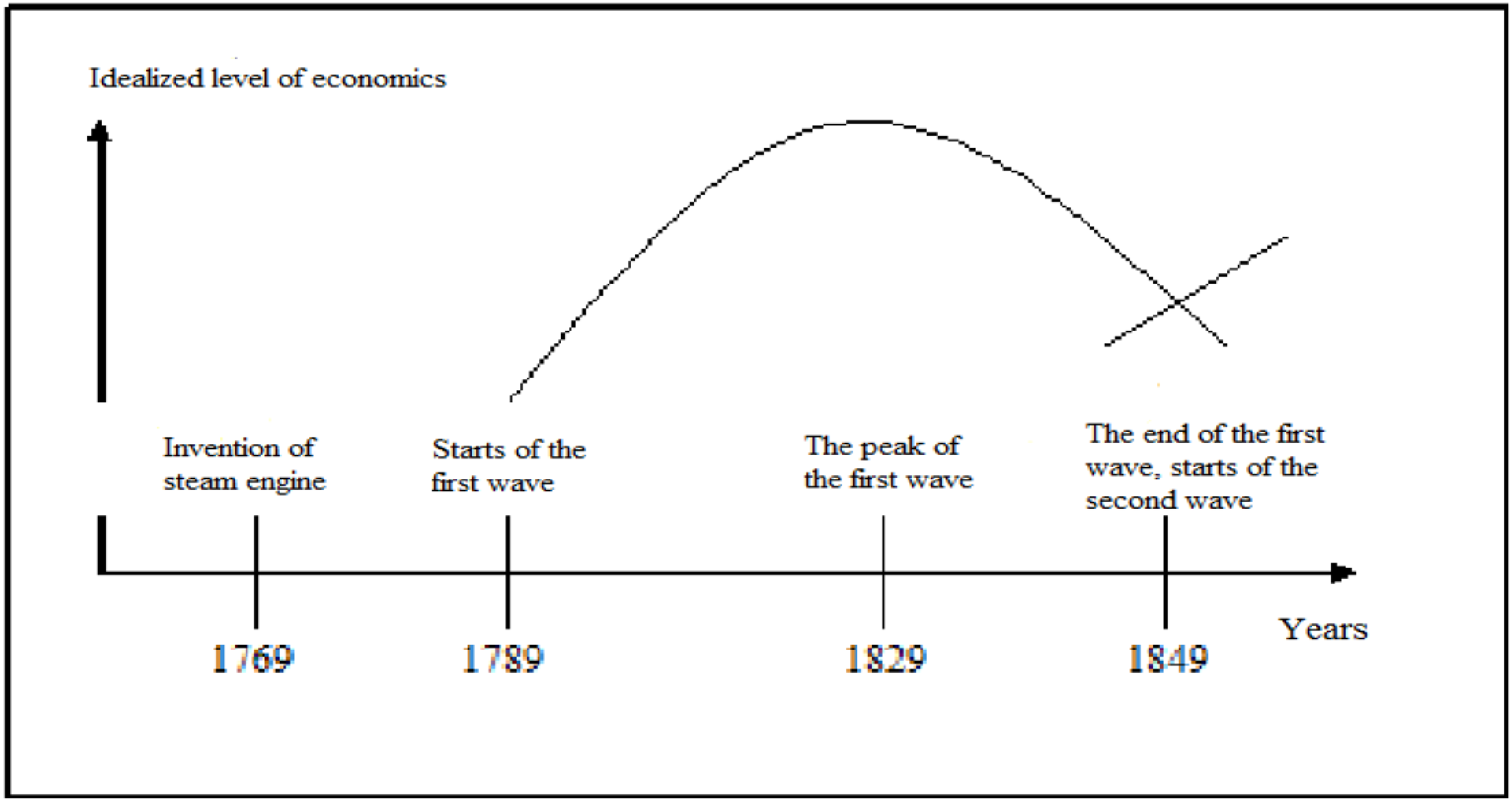
Recovery (AB segment), elevation (BC), loss (CD) and recession (DA). One life-cycle of the long-wave is measured to last on average 54 years.

Idealized long cycle

Structure of the long wave

1. The figure shows an idealized long wave (Kondratieff cycle) representing long-term economic growth over time.
2. The vertical axis represents the GDP growth rate, while the horizontal axis represents time (years).
3. One complete long wave has an average duration of about 54 years, as indicated on the time axis.
4. Point A (trough) represents the lowest phase of the economy, characterized by recession or depression.
5. The AB segment (Recovery phase) shows gradual economic revival, where investment, employment, and output begin to rise.
6. The BC segment (Elevation / Expansion phase) represents rapid economic growth driven by innovation, high productivity, and strong demand.
7. Point C is the peak of the long wave, where GDP growth reaches its maximum level.
8. The CD segment (Loss / Slowdown phase) indicates declining growth due to market saturation, rising costs, and reduced profitability.
9. The DA segment (Recession phase) shows economic contraction, falling investment, and rising unemployment, leading back to a trough.
10. After recession, the economy prepares for a new long wave, often driven by new technological innovations replacing older ones.

The first long cycle (1789-1849)



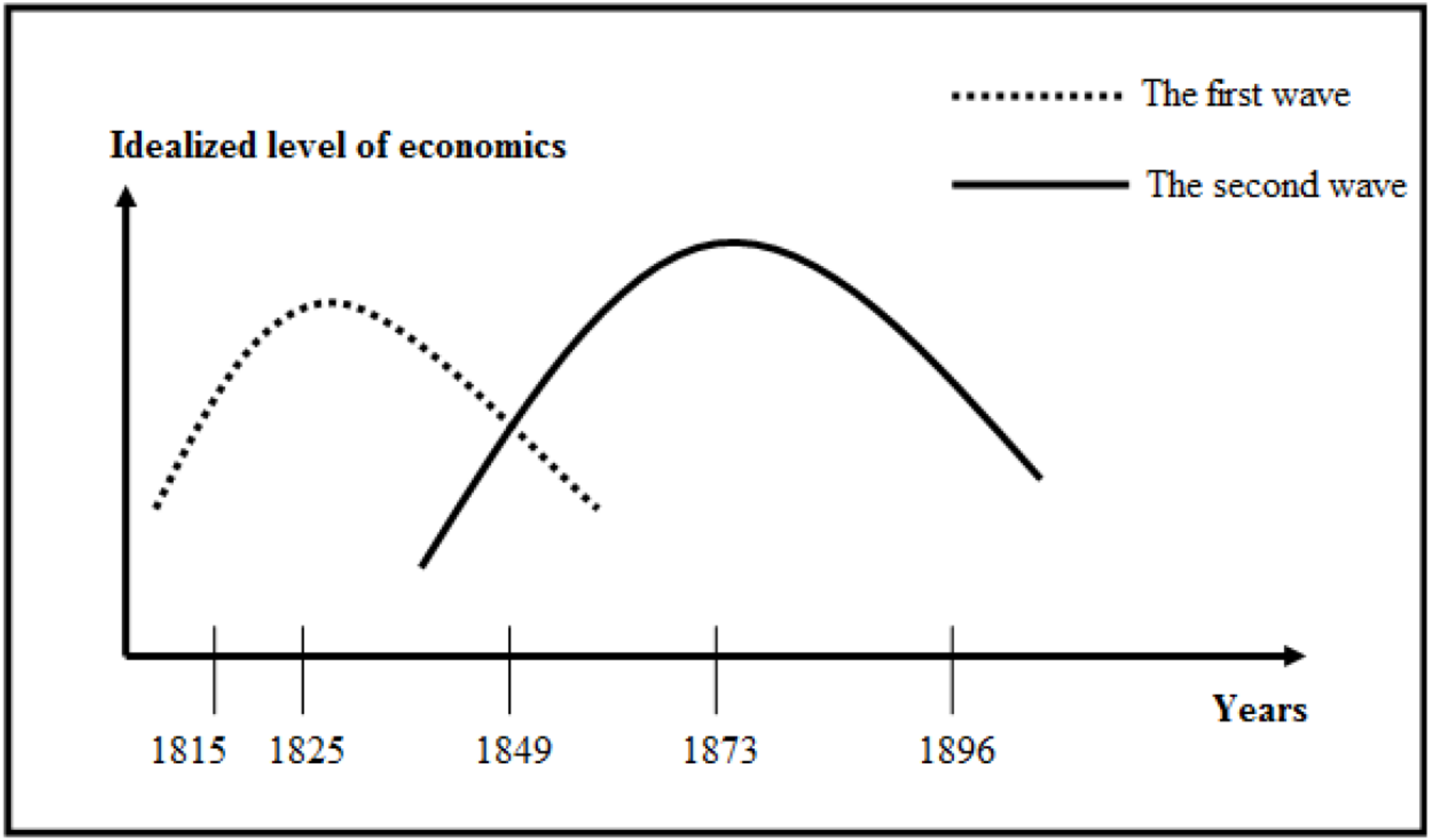
The first long cycle (1789-1849)

1. The **first long economic cycle** formally begins around **1789 (French Revolution)**, but its **technological roots date back to 1769** with **James Watt's steam engine**.
2. The **steam engine** was the **core technological driver** of the Industrial Revolution and the **main cause of the first Kondratieff long wave**.
3. The **Industrial Revolution** marked a transition from a **farm-based (agrarian) economy** to a **factory-based industrial economy**.
4. This transformation initially occurred in **England**, then spread to **Western Europe and North America** at different times.
5. The revolution led to a **significant improvement in living standards** for large sections of society over the long term.
6. Before industrialization, **artisans and manual craftsmen** produced goods in small manufactories with limited productivity.
7. Key textile innovations included: **Flying shuttle (John Kay, 1733)**, **Spinning jenny (James Hargreaves, 1766)**, **Water frame (Richard Arkwright, 1770)**
8. The application of the **steam engine to spinning machines (1790)** removed dependence on water power and **eliminated seasonal production limits**.
9. **Mechanical handwork was replaced by mass production**, marking the birth of the modern factory system.
10. Rapid growth of textile factories increased demand for **stronger machinery**, leading to the expansion of the **iron, steel, and cast-iron industries**.

The first long cycle (1789-1849)

1. Industrial concentration increased as steam power could be transmitted by belts and shafts, allowing factories to cluster together.
2. Factories were located near coal and iron mines, ports, and urban centers, reducing transport costs and ensuring labor availability.
3. Industrialization generated large-scale employment, reducing unemployment and increasing overall consumption demand.
4. Rural populations migrated to cities, accelerating urbanization and the growth of working-class settlements.
5. Factory owners aimed to minimize costs and maximize profits, which encouraged continuous technological improvement.
6. This competitive environment gave rise to modern industrial competition and increased investment in research and development (R&D).
7. By 1830, England produced nearly one-third of the world's manufactured goods, becoming the global industrial leader.
8. European dominance expanded through technological superiority, global trade, capital export, and colonization, laying foundations for imperialism.
9. Industrial growth also had negative phases: climate shocks and poor harvests raised food prices, reduced industrial demand, and caused unemployment.
10. Governments responded by promoting agricultural mechanization (reapers, threshers, fertilizers), improving food supply and stabilizing living standards.

The second long cycle (1849-1896)



The second wave was caused by usage of the locomotive

The second long cycle (1849-1896)

1. The 19th century is widely regarded as the age of capitalist prosperity, marked by rapid industrial and economic expansion.
2. During the early 19th century, the recession phase of the first long wave overlapped with the rise of the second long wave.
3. Kondratieff identified the first long-term economic decline occurring during this transition period.
4. The roots of the second long wave emerged during the downturn of the first wave, consistent with Kondratieff's theory.
5. The railway revolution was the main technological driver of the second long wave.
6. Although the locomotive was invented in 1815, it became economically transformative only after widespread railway construction.
7. The first steam railway line (1825, Great Britain) is considered the formal beginning of the second long wave.
8. Large-scale railway development required time for rail lines, tunnels, bridges, and stations, delaying full economic impact.
9. Railway construction generated massive employment and revived metallurgy and metal-processing industries.
10. By 1850, railway networks in the United States exceeded 9,000 km, enabling rapid inland expansion..

The second long cycle (1849-1896)

1. European migration to North America accelerated railway growth as settlers brought technical skills and capital from Britain.
2. The second long wave in the U.S. was strongly influenced by British technological leadership.
3. Railroads spread across Europe, Russia, India, and South America, transforming land transportation.
4. Railways reduced transport costs, increased speed, eliminated regional isolation, and expanded domestic markets.
5. Steamships, first introduced in 1807, further strengthened the second wave by improving water transport.
6. Advances such as steel hulls and propellers improved steamship efficiency and reduced operating costs.
7. Continuous technological innovation across Europe, North America, and Japan sustained long-term economic growth.
8. The demographic explosion of the 19th century played a key role by expanding markets and profit opportunities.
9. Population growth encouraged capital investment, urban construction, banking expansion, and corporate consolidation.
10. According to Kondratieff, the turning point of the second long wave occurred in 1873, followed by decline until new innovations emerged around 1896..

The third cycle (1896 – 1945)

1. Electricity, discovered during the peak of the second long wave, became the initiator of the third long wave.
2. Electricity could not drive the second wave because it was not widely applied until after 1886.
3. Joseph A. Schumpeter identified electrification as the key force behind the take-off of the third wave.
4. The third long wave was most prominent in the United States, England, and France.
5. Electrification replaced belt-driven power systems, which were inefficient and limited energy transfer.
6. Electric power allowed direct and efficient conversion of energy into mechanical motion.
7. Electricity enabled the spread of modern lighting systems, improving factory working conditions.
8. A major breakthrough of electrification was the introduction of conveyor-based production systems.
9. The conveyor system, pioneered by Henry Ford, fully mechanized industrial production.
10. Mass production of cars, engines, motorcycles, tractors, and airplanes became characteristic of the third wave.

The third cycle (1896 – 1945)

1. Conveyor-based manufacturing greatly increased output, reduced costs, and improved productivity.
2. Industries required less labor and raw material, leading to lower waste, losses, and pollution.
3. These mass-production techniques also transformed agriculture, making it more mechanized and efficient.
4. The widespread use of tractors, combine harvesters, and fertilizers reduced the risk of global famine.
5. World War I and the Russian October Revolution strongly influenced the take-off phase of the third wave.
6. During wartime, industries focused on military production, delaying adaptation to civilian needs after the war.
7. Kondratieff identified the 1920s as the upper turning point of the third long wave.
8. The Great Depression (1929–1933) marked the beginning of the recession phase of the third wave.
9. The crisis increased the role of state intervention, leading to policies such as the New Deal in the U.S.
10. The third long wave reached its lowest point during World War II (1939–1945), after which it transitioned into the fourth long wave.

The fourth wave (1945 – 1995)

1. World War II caused massive human losses, killing millions and radically transforming the global geopolitical and social order.
2. Since no major battles occurred on U.S. soil, the United States emerged economically stronger than other industrial nations.
3. The U.S. became the main donor for rebuilding post-war Europe, while also gaining returns through military and industrial supplies.
4. Over two decades after WWII, the U.S. provided approximately 97 billion dollars in aid to the rest of the world.
5. The U.S. dollar became the dominant international currency, strengthening American economic leadership.
6. European countries and Japan accumulated large amounts of U.S. capital, fostering the rise of powerful transnational corporations.
7. The United States maintained global economic dominance by supplying loans to war-damaged countries, ensuring demand for American goods.
8. The automobile industry became the centerpiece of post-war economic growth and mass consumption.
9. Car manufacturing was highly receptive to technological and scientific advancements in production.
10. Post-war industrial growth required a highly educated and skilled workforce, especially engineers and technicians.

The fifth long wave (1995 – 2030?)

1. The global economy is currently experiencing the fifth long wave, driven by an unprecedented abundance of innovations.
2. This wave may be shorter than previous long waves due to the rapid pace of technological change.
3. The fifth wave began with the widespread adoption of information and communication technologies (ICT).
4. This period marks the transition from the industrial age to the information age.
5. Modern economies are increasingly based on the creation, processing, and use of information rather than physical production.
6. Telecommunications and IT have caused major structural changes, intensifying globalization.
7. Proponents argue that the IT revolution is more transformative than any previous technological shift.
8. Technological diffusion has accelerated dramatically compared to earlier innovations.
9. For example, radio reached 50 million users in 38 years, whereas the internet reached 50 million users in just 4 years.
10. By 2011, over 2 billion people worldwide were using the internet..

The fifth long wave (1995 – 2030?)

1. This rapid diffusion shows that information exchange now grows at a geometric (exponential) rate.
2. Traditional long-term economic forecasting has become difficult due to high market volatility and rapid innovation cycles.
3. As a result, economists can now predict only broad trends rather than precise long-term outcomes.
4. Evidence suggests that the peak of the fifth long wave may already have occurred.
5. The recent global financial crisis, triggered by the collapse of the U.S. banking system, slowed the economic take-off.
6. The growth phase of the fifth wave was shorter than earlier waves due to fast adoption of new technologies.
7. The computer was the key innovation initiating the fifth long wave, with rapidly declining production costs.
8. Computer applications expanded from military and scientific uses to everyday personal and business activities.
9. Emerging technologies include artificial intelligence and expert systems, where computers learn from experience and apply logical reasoning.
10. Advances in biotechnology and biochemistry integrated with IT are expected to trigger the sixth long wave.

The sixth long wave (2030? - ?)

1. It is difficult to predict how the world will look like in twenty years especially during the high pace of technological developments.
2. According to the trend the sixth wave is expected to begin after ??.
3. As people increasingly will take more care of their health and well-being, continual exploration of longer life age is predicted to be the driving force of the sixth wave.
4. Developments in neurology will enable brains transplantation with available memory (skills).
5. Therefore people minds are expected to remain active for longer periods.
6. There will be attempts to explore the parts of the brain that are in charge of emotion determination, cognition, and sensation.
7. The cells are expected to be examined in the nano-molecular level (The sixth Kondratieff – long waves of prosperity). Findings retrieved will be implemented in the innovative industries.
8. Neuron technologies have potential to create new industries as well as new forms of social organizations. Nano-technologies will be essential at this period.
9. The sixth wave will be replaced by the seventh. Furthermore, it might be possible to break each of the long-cycle observed by Kondratieff. However, it is not possible predict future scenarios accurately.

Long Wave	Approx. Period	Core Global Technology	Indian Examples / Applications
First Long Wave	~1780–1849	Steam power, mechanized textiles	• Introduction of **steam-powered textile mills** in Bombay (Mumbai) and Ahmedabad (mid-19th century) • **Coal mining** in Raniganj coalfields (1774 onward) • Early mechanized **cotton spinning and weaving**
Second Long Wave	~1850–1896	Railways, steel, steamships	• **Indian Railways** (first railway line Bombay–Thane, 1853) • Expansion of **railway workshops** (Jamshidpur, Howrah) • Growth of **iron and steel demand** leading to Tata Iron & Steel Company (later TISCO, 1907 – roots in this wave)
Third Long Wave	~1896–1945	Electricity, chemicals, mass production	• **Hydroelectric power projects** (Shivanasamudra, 1902) • Electrification of cities (Calcutta, Bombay, Madras) • Establishment of **chemical, cement, and fertilizer industries** • Early automobile assembly and engineering workshops
Fourth Long Wave	~1945–1990	Oil, automobiles, consumer durables, mass manufacturing	• **Public sector industrialization** (steel plants: Bhilai, Rourkela, Durgapur) • Expansion of **petroleum refining** (IOC, BPCL, HPCL) • Growth of **automobile industry** (Hindustan Motors, later Maruti) • Large-scale **power plants and heavy engineering (BHEL)**
Fifth Long Wave	~1990–present	Information Technology, telecommunications, digital economy	• **IT and software industry** (TCS, Infosys, Wipro) • **Telecom revolution** (mobile phones, internet penetration) • **Digital India**, e-governance, fintech (UPI, Aadhaar) • Growth of **start-ups, AI, data analytics, and cloud computing**

Sixth Long Wave Technologies in India??

Technology Domain	What Defines the Sixth Wave	India-Specific Forecast & Examples
Clean & Renewable Energy	Decarbonization, net-zero systems	• Large-scale **solar & wind** (hybrid parks) • **Green hydrogen** (steel, fertilizer, transport) • **Grid-scale storage** (Li-ion, sodium-ion)
Electric & Smart Mobility	Electrification + autonomy	• **EVs** • **Battery manufacturing** • **Smart charging infrastructure**
Artificial Intelligence (AI)	Knowledge automation, learning systems	• **AI in governance** (health, land records) • **AI for manufacturing** (predictive maintenance) • **Indigenous AI models** (Indian languages)
Biotechnology & Bio-manufacturing	Biology as production platform	• **Vaccines & biopharma** leadership • **Synthetic biology** (enzymes, biofuels) • **Agri-biotech** (climate-resilient crops)
Quantum Technologies	New computation & sensing paradigms	• **Quantum computing** • **Quantum sensors** (defence, navigation)
Advanced Materials & Nanotech	High performance, low footprint	• **Composites** for aerospace/Evs • **Graphene & nano-coatings** (energy, water)
Circular Economy & Green Industry	Resource efficiency	• **Recycling & waste-to-value** • **Low-carbon cement & steel**
Digital Public Infrastructure (DPI)	Platform-led inclusion	• **UPI-like stacks** for health, logistics • **AI-enabled DPI** for service delivery
Agri-tech & Food Systems	Precision, resilience	• **Precision farming** (IoT, drones) • **Alternative proteins** & cold chains
Space & Dual-use Tech	Downstream applications	• **Small satellites**, EO analytics • **Private space ecosystem**

Long Wave Mining Technologies in India

Long Wave	Approx. Period	Global Mining Technology Driver	Examples in Indian Mining Context
First Long Wave	~1780–1849	Steam power, mechanization	• Steam-powered pumping & winding in coal mines (Raniganj coalfields) • Early mechanized coal extraction replacing manual mining • Use of steam engines for dewatering deep mines
Second Long Wave	~1850–1896	Railways, steel, steam haulage	• Mine railways & tramways for coal and ore transport • Expansion of iron ore & coal mining to supply railways • Steel demand leading to systematic iron ore mining in Odisha & Jharkhand
Third Long Wave	~1896–1945	Electricity, conveyors, mass production	• Electrification of underground coal mines • Introduction of belt conveyors for mineral transport • Electric pumps, fans, and hoists improving safety and productivity
Fourth Long Wave	~1945–1990	Oil-based machinery, heavy earthmoving equipment	• Large-scale opencast mining using draglines, shovels, dumpers Mechanization in Coal India mines • Diesel-powered hauling systems
Fifth Long Wave	~1990–present	Digitalization, automation, ICT	• GPS-based mine surveying • Mine planning software (Surpac, MineScape) • Remote sensing & GIS for mineral exploration • Real-time production & safety monitoring

What's next in Mining?

Technology Domain	Sixth Long Wave Focus	Forecast for India (with Mining Integration)
Clean & Renewable Energy	Net-zero, decarbonized energy systems	• Large-scale solar & wind parks • Green hydrogen for steel & mining operations • Renewable-powered mine electrification
Electric & Smart Mobility	Electrification & automation	• Electric mine haul trucks • EVs for mineral logistics • Battery swapping for mining fleets
Artificial Intelligence (AI)	Knowledge automation	• AI-based mineral exploration • Predictive maintenance of mining equipment • AI-driven mine planning & scheduling
Digital Mining (Industry 5.0)	Human–machine collaboration	• Smart mines (IoT sensors, digital twins) • Real-time monitoring of slope stability & ventilation • Autonomous drilling & blasting
Critical Minerals & Energy Metals	Resource security	• Exploration & mining of lithium, cobalt, nickel, REEs • Deep-sea & unconventional mineral exploration • Strategic mineral stockpiling
Advanced Materials & Nanotechnology	High-performance materials	• Nanomaterials for mineral processing efficiency • Lightweight composites for mining machinery
Circular Economy & Urban Mining	Resource reuse	• Metal recycling (aluminum, copper, lithium) • Recovery of minerals from e-waste • Mine waste reutilization
Carbon Capture & Storage (CCS)	Climate mitigation	• CO_2 storage in abandoned coal mines • Carbon-neutral mining clusters
Robotics & Automation	Safety & productivity	• Robotic mining in hazardous zones • Reduced human exposure underground
Space & Remote Sensing	Earth observation	• Satellite-based mineral prospecting • Illegal mining detection
Skill & Knowledge Economy	Human capital	• New roles: Mining data scientists, AI mine engineers • High-skill employment generation

Mine backfill technologies

Backfill in underground metalliferous mines

Uncemented backfill (before 1950s)

- Hydraulic slurry backfill (Sand, crushed waste rock etc.)
- Rock fill
- Sand fill
- Aggregate fill

Cemented backfill (after 1950s)

- Cemented hydraulic slurry backfill (CHSB): 1950s
- Cemented rock fill (CRF): 1960s
- Cemented aggregate fill (CAF): 1970s
- Paste backfill: 1980s